

Compile, Run, help command, exit program:

```
((base) matthewbosworth@Matthews-Laptop-4 OS-Group-Assignment1-C % nano dos_interpreter.c
((base) matthewbosworth@Matthews-Laptop-4 OS-Group-Assignment1-C % make
gcc -Wall -Wextra -std=c99 -o dos_interpreter dos_interpreter.c
((base) matthewbosworth@Matthews-Laptop-4 OS-Group-Assignment1-C % ./dos_interpreter
=====
  DOS Command Interpreter in C
=====
Type 'help' for available commands
Type Ctrl-C to exit
=====

DOS> help

Before command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

Available DOS Commands:
cd <directory>           - Change directory
dir [directory]          - List directory contents
type <filename>          - Display file contents
del <filename>           - Delete a file
ren <old_name> <new_name> - Rename a file
copy <source> <destination> - Copy a file
pwd                      - Show current directory
help                    - Show this help message

After command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

DOS> ^C
(base) matthewbosworth@Matthews-Laptop-4 OS-Group-Assignment1-C %
```

Run test setup for showcasing commands with instructions on how to test them:

```
((base) matthewbosworth@Matthews-Laptop-4 OS-Group-Assignment1-C % ./setup_test.sh
Setting up test environment for DOS Command Interpreter...

Test environment created in test_demo/

Directory structure:
find: test_demo/: No such file or directory

You can now test the following scenarios:

1. cd command:
DOS> cd test_demo
DOS> pwd
DOS> cd ..

2. dir command:
DOS> dir
DOS> dir test_demo

3. type command:
DOS> type test_demo/testfile1.txt

4. copy command:
DOS> copy test_demo/source.txt test_demo/destination.txt
DOS> dir test_demo

5. ren command:
DOS> ren test_demo/oldname.txt test_demo/newname.txt
DOS> dir test_demo

6. del command:
DOS> del test_demo/todelete.txt
DOS> dir test_demo

7. Error cases:
DOS> cd
DOS> copy onefile.txt
DOS> cd too many arguments

Setup complete! Run ./dos_interpreter to start testing.
((base) matthewbosworth@Matthews-Laptop-4 OS-Group-Assignment1-C % ls
dos_interpreter      dos_interpreter.c    Makefile             setup_test.sh        test_demo
((base) matthewbosworth@Matthews-Laptop-4 OS-Group-Assignment1-C % cd test_demo
((base) matthewbosworth@Matthews-Laptop-4 test_demo % ls
oldname.txt  source.txt  subdir1    subdir2    testfile1.txt  testfile2.txt  todelete.txt
```

Commands: cd, pwd (not required), dir

```
(base) matthewbosworth@Matthews-Laptop-4 OS-Group-Assignment1-C % ./dos_interpreter
=====
DOS Command Interpreter in C
=====
Type 'help' for available commands
Type Ctrl-C to exit
=====

DOS> cd test_demp

Before command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

Error changing directory: No such file or directory

After command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

DOS> cd test_demo

Before command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

Directory changed successfully.

After command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C/test_demo

DOS> pwd

Before command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C/test_demo

Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C/test_demo

After command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C/test_demo

DOS> cd ..

Before command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C/test_demo

Directory changed successfully.

After command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

DOS> dir

Before command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

total 120
drwxr-xr-x@ 8 matthewbosworth  staff   256 Feb 10 18:16 .
drwx-----@ 15 matthewbosworth  staff   480 Feb 10 11:11 ..
-rw-r--r--@ 1 matthewbosworth  staff  6148 Feb 10 18:16 .DS_Store
-rwxr-xr-x@ 1 matthewbosworth  staff 34396 Feb 10 17:59 dos_interpreter
-rw-r--r--@ 1 matthewbosworth  staff  7777 Feb 10 11:11 dos_interpreter.c
-rw-r--r--@ 1 matthewbosworth  staff   288 Feb  8 13:39 Makefile
-rwxr-xr-x@ 1 matthewbosworth  staff  1593 Feb 10 18:15 setup_test.sh
drwxr-xr-x@ 10 matthewbosworth  staff   320 Feb 10 18:16 test_demo

After command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C
```

Commands: dir (continued), type, copy

```
DOS> dir test_demo

Before command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

total 56
drwxr-xr-x@ 10 matthewbosworth  staff   320 Feb 10 18:16 .
drwxr-xr-x@  8 matthewbosworth  staff   256 Feb 10 18:16 ..
-rw-r--r--@  1 matthewbosworth  staff  6148 Feb 10 18:16 .DS_Store
-rw-r--r--@  1 matthewbosworth  staff    19 Feb 10 18:16 oldname.txt
-rw-r--r--@  1 matthewbosworth  staff    27 Feb 10 18:16 source.txt
drwxr-xr-x@  3 matthewbosworth  staff    96 Feb 10 18:16 subdir1
drwxr-xr-x@  2 matthewbosworth  staff    64 Feb 10 18:16 subdir2
-rw-r--r--@  1 matthewbosworth  staff    20 Feb 10 18:16 testfile1.txt
-rw-r--r--@  1 matthewbosworth  staff    20 Feb 10 18:16 testfile2.txt
-rw-r--r--@  1 matthewbosworth  staff    19 Feb 10 18:16 todelete.txt

After command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

DOS> type test_demo/testfile1.txt

Before command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

This is test file 1

After command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

DOS> copy test_demo/source.txt test_demo/destination.txt

Before command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

File copied successfully.

After command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C
```

Continued confirmation that copy command works properly:

```
(base) matthewbosworth@Matthews-Laptop-4 OS-Group-Assignment1-C % ./dos_interpreter
=====
DOS Command Interpreter in C
=====
Type 'help' for available commands
Type Ctrl-C to exit
=====

DOS> type test_demo/source.txt

Before command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

Sample content for copying

After command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

DOS> type test_demo/destination.txt

Before command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

Sample content for copying

After command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

DOS> dir test_demo

Before command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

total 64
drwxr-xr-x@ 11 matthewbosworth  staff   352 Feb 10 18:20 .
drwxr-xr-x@  8 matthewbosworth  staff   256 Feb 10 18:32 ..
-rw-r--r--@  1 matthewbosworth  staff  6148 Feb 10 18:16 .DS_Store
-rw-r--r--@  1 matthewbosworth  staff    27 Feb 10 18:22 destination.txt
-rw-r--r--@  1 matthewbosworth  staff    19 Feb 10 18:16 oldname.txt
-rw-r--r--@  1 matthewbosworth  staff    27 Feb 10 18:16 source.txt
drwxr-xr-x@  3 matthewbosworth  staff    96 Feb 10 18:16 subdir1
drwxr-xr-x@  2 matthewbosworth  staff    64 Feb 10 18:16 subdir2
-rw-r--r--@  1 matthewbosworth  staff   20 Feb 10 18:16 testfile1.txt
-rw-r--r--@  1 matthewbosworth  staff   20 Feb 10 18:16 testfile2.txt
-rw-r--r--@  1 matthewbosworth  staff    19 Feb 10 18:16 todelete.txt

After command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

DOS> █
```

Commands ren and del:

```
DOS> ren test_demo/oldname.txt test_demo/newname.txt

Before command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

File renamed successfully.

After command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

DOS> dir test_demo

Before command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

total 64
drwxr-xr-x@ 11 matthewbosworth  staff   352 Feb 10 18:40 .
drwxr-xr-x@  8 matthewbosworth  staff   256 Feb 10 18:32 ..
-rw-r--r--@  1 matthewbosworth  staff  6148 Feb 10 18:16 .DS_Store
-rw-r--r--@  1 matthewbosworth  staff    27 Feb 10 18:22 destination.txt
-rw-r--r--@  1 matthewbosworth  staff    19 Feb 10 18:16 newname.txt
-rw-r--r--@  1 matthewbosworth  staff    27 Feb 10 18:16 source.txt
drwxr-xr-x@  3 matthewbosworth  staff    96 Feb 10 18:16 subdir1
drwxr-xr-x@  2 matthewbosworth  staff    64 Feb 10 18:16 subdir2
-rw-r--r--@  1 matthewbosworth  staff    20 Feb 10 18:16 testfile1.txt
-rw-r--r--@  1 matthewbosworth  staff    20 Feb 10 18:16 testfile2.txt
-rw-r--r--@  1 matthewbosworth  staff    19 Feb 10 18:16 todelete.txt

After command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

DOS> del test_demo/todelete.txt

Before command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

File deleted successfully.

After command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

DOS> dir test_demo

Before command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

total 56
drwxr-xr-x@ 10 matthewbosworth  staff   320 Feb 10 18:40 .
drwxr-xr-x@  8 matthewbosworth  staff   256 Feb 10 18:32 ..
-rw-r--r--@  1 matthewbosworth  staff  6148 Feb 10 18:16 .DS_Store
-rw-r--r--@  1 matthewbosworth  staff    27 Feb 10 18:22 destination.txt
-rw-r--r--@  1 matthewbosworth  staff    19 Feb 10 18:16 newname.txt
-rw-r--r--@  1 matthewbosworth  staff    27 Feb 10 18:16 source.txt
drwxr-xr-x@  3 matthewbosworth  staff    96 Feb 10 18:16 subdir1
drwxr-xr-x@  2 matthewbosworth  staff    64 Feb 10 18:16 subdir2
-rw-r--r--@  1 matthewbosworth  staff    20 Feb 10 18:16 testfile1.txt
-rw-r--r--@  1 matthewbosworth  staff    20 Feb 10 18:16 testfile2.txt

After command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

DOS> █
```

Error cases:

```
DOS> cd

Before command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

Error: Too few arguments. Usage: cd <directory>

After command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

DOS> copy onefile.txt

Before command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

Error: Too few arguments. Usage: copy <source> <destination>

After command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

DOS> cd too many arguemnets hello

Before command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

Error: Too many arguments. Usage: cd <directory>

After command execution:
Current directory: /Users/matthewbosworth/Desktop/OS-Group-Assignment1-C

DOS> █
```